Sahar Levy

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Professional Summary

U.S. citizen and mechanical engineering student at Cal Poly Pomona with hands-on experience in CAD modeling, mechanism design, additive manufacturing, and shop fabrication. Contributor to multidisciplinary engineering teams spanning unmanned aircraft, rocket materials research, and award-winning competition design. Skilled at taking mechanical systems from concept through design, prototyping, and testing, with a strong interest in materials engineering and energy engineering. Recognized with the Gene Haas Foundation Scholarship for technical aptitude and leadership.

Education

California State Polytechnic University, Pomona
B.S., Mechanical Engineering. Fall 2025 - Present. GPA: 3.43. Expected Graduation 2029

Irvine Valley College
Design, Model Making and Rapid Prototyping Classes. Spring 2024 - Summer 2025

Woodbridge High School, Irvine
High School Diploma. GPA: 4.18. Graduated 2025

Experience

IAM3D Competition Team Member, 2025 - Present*, Cal Poly Pomona*

- Design and optimize land rover chassis structures in SolidWorks, balancing strength, weight, and mission requirements.
- Apply DFMA and DFAM principles to prototype and iterate on critical components using thermoplastics.
- Earned first place at the ASME eFX Dallas competition out of 20 national teams.

Liquid Rocket Lab Team Member, 2026 - Present*, Cal Poly Pomona*

- Verify mechanical properties of metal 3D-printed parts against published values after prior teams could not replicate them, intended for future launch vehicle and rocket engine use.
- Model test specimens in Fusion 360 and develop test procedures following ASTM standard E8.
- Run tensile tests on specimens and write twice-yearly reports to be delivered to sponsors.

Banshee UAV Team Member, 2025 - Present, *Cal Poly Pomona*

- Fabricate and assemble VTOL UAV components including XPS foam wing cutting and 3D printing.
- Coordinate across aerodynamics, avionics, design, flight testing, and robotics subteams to verify structural integrity and operational readiness.
- Team sponsored by Lockheed Martin, Southern California Edison, Air Force Research Laboratory, Robotis, and General Atomics.

Activities & Leadership

American Society of Mechanical Engineers (ASME), 2025 - Present, *Cal Poly Pomona*
Engage with the professional engineering community through industry speaker events and workshops.

SkillsUSA Executive Member, 2025, *Woodbridge High School*
Led and organized chapter events and competition preparations, and promoted STEM collaboration and career readiness.

COSMOS Program Participant, Summer 2024*, University of California, Irvine*
A highly selective four-week University of California program in collaboration with UCI faculty, culminating in a team project analyzing hybrid-series aircraft propulsion with a symposium presentation.

Scouts Counselor, 2020 - 2025, *Orange County Jewish Scouts*
Organized weekly programs and camps for youth groups, handling logistics and safety while mentoring younger members.

Awards & Honors

- 1st Place, ASME eFX Dallas Competition (2025)
- Cal Poly Pomona Dean's List (2025)
- Gene Haas Foundation Scholarship Recipient (2025)
- SkillsUSA Gold Chapter of Distinction (2025)
- SkillsUSA Automated Manufacturing Technology State Qualifier (2025)
- Presidential Volunteer Service Award (2023, 2024)

Skills

Technical: CAD Modeling (SolidWorks, Fusion 360), MATLAB, FDM and Metal Additive Manufacturing / 3D Printing, Structural Design, Rapid Prototyping, Manual Machining, Metal Casting, Shop Experience